SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*
Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Description about the assessment tool:
Short-answer questions effectively measure conceptual understanding, logical reasoning, and the ability to apply theoretical knowledge without requiring lengthy descriptive responses. This assessment method is also efficient for evaluating multiple OSI-related concepts within a limited time while ensuring objective and consistent evaluation.

Code of Conduct:

I D Naveen Kumar (192511401) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

D. Naveen Kumar
Signature of the student:

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Understanding of Concepts	25%	2	2	2	2
Explanation	25%	2	2	2	2
Application	20%	2	2	2	2
Organization	15%	1½	1½	1½	1½
Accuracy	10%	1	1	1	1
Clarity	5%	½	½	½	½

Signature of the Faculty

Total Marks 36 **/40**

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7. An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8. A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9. A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and

the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10. A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Given :-

* Original file size = 900MB
* Compression = 40%
* Encryption overhead = 5%

FORMULA

* Compressed Size = Original size × (1 - Compression %)
* Encrypted Size = Compressed Size × (1 + Encryption Overhead %)
* Percentage Reduction = $\frac{\text{Original size} - \text{Encrypted Size}}{\text{Original size}} \times 100$

CALCULATION

Compressed file size = 900 × (1 - 0.40)
= 540MB

Encryption Overhead = 5% of 540
= 27MB

Encrypted file Size = 540 + 27
= 567MB

Percentage Reduction = ((900 - 567) ÷ 900) × 100

= 37 %.

Final Answer:

Compressed File Size = 540 MB

Encrypted File Size = 567 MB

Overall Reduction = 37%.

APPLICATION

The presentation Layer is widely Used in

* HTTPS Secure Web Communication
* Cloud Storage Services
* Online banking
* Video conferencing
* Secure file transfer Systems.

CONCLUSION:

By combining compression and encryption, the presentation layer improves both network efficiency and data security, making communication faster and more secure.

7.

Given:-

* File Size = 3GB = 3000MB (Engineering convention)
* Request Size = 5MB
* Throughput = 120Mbps

FORMULA:

Number of Requests = Total Data ÷ Request Size

Transmission time = Total Data (in Mb) ÷ Throughput

CALCULATION:

Number of Requests = 3000 ÷ 3
= 1000 Requests

Convert Data into Megabits 3000 MB × 8

= 24000 Mb

Transmission time = 24000 ÷ 120

= 200 Seconds

Final Answer

* Total Requests = 1000

* Transmission time = 200 Seconds (≃ 3.33 minutes)

Application:

Application Layer protocols are used in:

* FTP file transfer

* Web browsing

* Email Communication.

* Cloud Storage
* Remote Server Management

Conclusion:

The Application Layer simplifies communication between Users and network services while ensuring efficient and reliable data transfer.

8.

Given :-

Application Delay = 4ms

Transport Delay = 3ms

Network Delay = 5ms

Data Link Delay = 2ms

Physical Delay = 1ms

Propagation Delay = 18 ms

Transmission Delay = 12ms

FORMULA

Total Delay = Processing Delay + Propagation Delay + Transmission Delay.

CALCULATION :

Processing Delay = 4 + 3 + 5 + 2 + 1
= 15ms

Total End - to - End Delay = 15 + 18 + 12
= 45 ms

Final Answer

Total End - to - End Delay = 45 ms

APPLICATION

Delay analysis is important in :

* Online gaming

Video Conferencing
Voice Over IP (VoIP)
* Cloud Computing
* Industrial IoT Systems.

CONCLUSION

Minimizing end-to-end delay improves network performance and enhances the quality of real time applications.

9.

Given :-

* Useful Data = 80 MB = 80 × 1024 × 1024
= 83,886,080 bytes

* Frame Size = 1600 bytes

* Overhead Per Frame = 80 bytes

FORMULA

Useful Data per frame = Frame Size - Overhead

Number of Frames = Total Useful Data ÷ Useful Data Per frame

Total overhead = Number of frames × Overhead Per frame

Transmission Efficiency = (Useful Data ÷ Total Transmitted Data) × 100

CALCULATION

Useful Data Per frame = 1600 - 80

= 1520 bytes

Number of Frames = 83,886,080 ÷ 1520

≈ 55,188 Frames

Total Overhead = 55,188 × 80

= 4,415,040 bytes

Total Data Transmitted = 83,886,080 + 4,415,040

= 88,301,120 bytes

Transmission Efficiency = (83,886,080 ÷ 88,301,120) x 100

≃ 95.0%

Final Answer

* Useful Data Per Frame = 1520 bytes
* Total Frames = 55,188
* Total Protocol Overhead = 4,415,040 bytes
* Transmission efficiency ≈ 95%

APPLICATION

Protocol overhead is considered when designing

* Ethernet networks
* wi-fi communication
* Cloud data centers
* internet backbone networks
* Enterprise LANs

CONCLUSION

Efficient Protocol design balances reliability and performance by minimizing overhead while ensuring accurate and secure data transmission.